

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
18 emerging paradigms in neuroscience—specifically classical electrody-
19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
20 tion, culminating in a topological fixed point: a unified “Self.” The
21 formalization is reported with its scope attached: the development
22 declares no axioms and every structure carrying physical content is
23 inhabited by a witness, while the two links that remain assump-
24 tions—that the field reaches the synchronized configuration, and that
25 a global section of the probability sheaf is the unity of experience—are
26 marked as such throughout rather than absorbed into the deriva-
27 tion. As a corollary, we show why rigid von Neumann architectures
28 (e.g., GPUs) are thermodynamically disadvantaged or structurally
29 constrained from reaching these specific continuous thermodynamic
30 attractors.

31 1 Introduction

32 The pursuit of a physical theory of consciousness typically involves search-
33 ing for the biological correlates of mind or treating subjective experience as
34 an abstract computational algorithm. This paper asks a more fundamental
35 question: If we assume only basic physical primitives—continuous symme-
36 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
37 of a unified, perceiving entity a natural physical attractor?

38 We propose a framework where the structural properties of consciousness

39 are derived as physical attractor states. Establishing a deductive chain from
 40 basic principles, we demonstrate how continuous symmetries break to form
 41 boundaries, how those boundaries dissipate free energy as non-equilibrium
 42 steady states, and how their evolution along paths of least action results
 43 in physical resonance with their environment. This eventually scales into
 44 the macroscopic electromagnetic unity proposed by emerging field theories
 45 of mind. To avoid critiques of “math theatre,” we delineate between literal
 46 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 47 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 48 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 49 spective aligned with Russellian Monism: objective measurements of macro-
 50 scopic electromagnetic fields constitute the third-person “outside,” while the
 51 system’s internal physical deformation to reach structural resonance consti-
 52 tutes the first-person intrinsic experience.

53 **2 Formalization in Lean 4**

54 A core contribution of this work is the verification of our deductive chain
 55 using the Lean 4 interactive theorem prover. This approach builds on recent
 56 efforts to formalize consciousness and brain semantics, such as the ‘OnlyOne’
 57 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
 58 2026). It is necessary to explicitly distinguish which claims have been math-
 59 ematically derived as theorems within the proof assistant, and which remain

irreducible physical postulates. Table 1 maps the primary claims of this paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Structural Resonance KL Bound	Gibbs’ inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (Phase3_KLBound)	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region’s stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a ThermodynamicCover , whose fields are now all local: the global section is constructed by probability_glue_unique and the former class field sync_to_section_eq is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14)	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly 1/2 without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (eq_of_avatar_eq), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4_RotatingFrame); identical frequencies	Theorem
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (kuramotoFieldlipschitz , kuramotoField_norm_le), and through every state at every instant passes exactly one trajectory, defined for all time (kuramoto_cauchy_problem). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (dynamic_potential_tendsto) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (dissipation_integral_tendsto). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (velocity_sq_tendsto_zero). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (kuramoto_tendsto_global_minimum), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (Examples §17)	Theorem (conditional)

Conceptual Claim	Lean 4 Implementation	Status
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (<code>Phase7.Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed.point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

62 2.1 Soundness of the formalization

63 Formalization is only as trustworthy as the axioms it rests on, and an axiom
64 that is inconsistent silently makes every theorem downstream of it vacuous
65 while leaving the build green. We report a case of exactly this, found by
66 auditing our own development, because the failure mode is easy to reproduce
67 and we have not seen it discussed in the applied-formalization literature.

68 Earlier drafts of this work declared five axioms. Three of them were
69 individually inconsistent: each, on its own, proves **False**. All three shared
70 one root cause — an axiom that constrains a symbol it does not itself bind.
71 Concretely:

- 72 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
73 energy functional as an *implicit* argument occurring only in its hypoth-
74 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
75 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields

76 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
77 this proves $1 = 0$.

- 78 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
79 a bath-entropy difference *for every instance* of the class. An instance
80 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 81 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P and Q , while σ
82 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
83 point mass against a uniform distribution refutes it.

84 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
85 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`
86 — and are refutable as soon as the probability presheaf admits two distinct
87 global sections. (Both were demoted to class fields at the time; both have
88 since been removed from the class altogether, for a separate reason given in
89 Derivation 5.)

90 The correction is uniform, and we state it as a rule we now follow: *a*
91 *physical postulate that mentions a class field must be a field of that class,*
92 *never a standalone axiom quantified over all instances.* A postulate should
93 be an obligation each model discharges, not a global claim about every model.
94 Recast this way, the development declares no axioms at all. Two of the five
95 became genuine theorems — pointwise vacuum minimization is now proved
96 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs' inequality was already one
97 — and the other three became instance obligations.

98 This makes non-vacuity the live question, since an uninhabitable class
 99 makes its theorems vacuous just as surely as an inconsistent axiom did.
 100 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with
 101 a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$;
 102 a perfectly resonant system discharging `kl_bound`; and a scalar field on
 103 a one-point spacetime discharging `ActionPrinciples`. Derivation 5's glu-
 104 ing structures are witnessed too, which was until recently the largest open
 105 gap: `Examples.lean` builds a `ThermodynamicCover` over a three-site sub-
 106 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 107 *overlapping* open patches whose intersection is a single shared site. The
 108 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 109 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 110 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 111 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 112 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 113 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked minimizes_potential`
 114 rather than assumed. `global_section_from_thermodynamics` then fires on
 115 this instance and glues the two patch-local sections into a unique global one.
 116 Two further covers answer the follow-up question of how non-degenerate such
 117 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 118 ing: the phase field can be made non-constant as a function, but not as a
 119 physical state, because `thermodynamic_equilibrium` forces every instance
 120 to a phase-locked configuration. `Examples.lean` §14 is built the other way

121 round, from two patches with independently chosen mass profiles that meet
 122 only on the site they share, and the section the gluing returns is neither of
 123 them. That is the same kind of result as `contracting_implies_const` in
 124 Derivation 6 — a proof that the witness is as strong as the class permits,
 125 rather than a witness that could be improved by trying harder.

126 What the witnesses establish is that the classes are inhabitable, so the
 127 theorems about them are not vacuous. What they do not establish is that
 128 a cortex is such an instance. In the cover above the phase field is already
 129 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 130 the equilibrium obligation dischargeable; getting a physical system into that
 131 configuration is the content of the informal dynamical argument, not of the
 132 Lean development.

133 The same audit turned up a defect of a different kind, worth record-
 134 ing because it is not an axiom problem and would not have been caught
 135 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
 136 crete edge weights converge to the continuous energy as the mesh is refined
 137 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 138 not merely unproved but false. Nothing required the edge regions to cover
 139 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 140 all empty satisfied the mesh hypothesis while contributing nothing to the
 141 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 142 quantified over a triangulation and then resolved the vertex type by instance
 143 search rather than through the bound variable, so the variable was never

144 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 145 gence is now stated along a *sequence* of triangulations of a named region
 146 S , each required to be regular — edge regions measurable, without self-
 147 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 148 plicial data in the sense that a vertex lies in the closure of every region
 149 of an edge at it. Closure rather than membership is forced: cells are dis-
 150 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 151 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 152 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 153 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 154 tion, reading the double sum as a Riemann sum would be a category error.
 155 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 156 the integrand, and convergence follows for uniformly continuous integrands
 157 as the mesh shrinks. The regularity conditions pull against each other, so
 158 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 159 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 160 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 161 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 162 the accompanying simulation measures; that would need a midpoint error
 163 term with a second derivative, which we have not developed.

164 A third defect was one of bookkeeping across files, and it is the kind that
 165 survives any number of green builds because each half is individually cor-
 166 rect. The development carried *two* Kuramoto potentials. Lyapunov descent

167 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 168 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 169 — phase-locked in both directions — and everything downstream of it, in-
 170 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 171 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 172 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 173 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 174 it has no minimum for phase-locking to attain, and the sentence “the phase-
 175 locked state minimizes the Lyapunov potential” was, read literally, false of
 176 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 177 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 178 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 179 frequency system — which is exactly the system Derivation 5 already worked
 180 with — the two potentials coincide there, and the descent result transports
 181 onto the coupling potential whose minimizers are the phase-locked states.
 182 Phase differences, phase-locking and the magnitude of the order parameter
 183 are all proved invariant under the change of frame, so nothing observable
 184 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 185 two oscillators with a common frequency, discharging every hypothesis. The
 186 restriction is real and we state it plainly: the reduction is exact only for
 187 identical frequencies. A genuine spread of natural frequencies leaves residual
 188 detunings in the reduced system, phase-locking then requires coupling above
 189 a critical K_c , and K_c is nowhere in our Lean development — it is studied

190 numerically, as Table 1 records.

191 A fourth defect was dimensional, and it concerned the one row of Table 1
192 that is still numerical rather than proved. `Phase8.ContinuousField.lean`
193 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
194 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
195 against an unnormalized measure carries the substrate’s mass twice over,
196 so a constant kernel K on a substrate of total mass m contributed Km^2 .
197 The comparison therefore depended on how large the substrate was rather
198 than on how strongly it was coupled, and any kernel whatever could be
199 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
200 ure concretely: on a two-site substrate carrying counting measure, a sys-
201 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
202 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
203 satisfied the old predicate. The fix is to require the substrate’s measure to
204 be a probability measure, which is the normalization in which the mean-
205 field derivation of $K_c = 2D$ is carried out; `mean.field.coupling.const`
206 then gives that a constant kernel has mean-field strength exactly K , and
207 `exhibits.phase.transition.const_iff` reduces the predicate to $K > 2D$.
208 We flag what this does and does not do. It makes the predicate say something
209 whose truth depends only on the physics, and §9 witnesses it on both sides of
210 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
211 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
212 at $K = 2D$, which is a separate question from whether the predicate is di-

213 mensionally sound. That bifurcation is proved below, in both directions,
 214 from an assumed stationary density; deriving the density itself needs the
 215 Fokker–Planck operator and is not done.

216 We draw no grand moral from this beyond the obvious one. The value
 217 of a proof assistant here was not that it certified our argument, but that it
 218 made a specific, checkable claim about what had been assumed — and that
 219 claim turned out to be false in a way no amount of reading the prose would
 220 have revealed.

221 3 Axiom 1: Symmetries and Phase Space

222 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 223 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
 224 Poincaré group) act upon field states, and the invariance of the system’s total
 225 energy under that action is recorded by the class `SymmetryInvariantAction`.
 226 Physically, it is such an invariance that Noether’s theorem converts into the
 227 conservation of energy and momentum. The scope of the *formal* claim is nar-
 228 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 229 a model must discharge, and no conserved quantity is ever constructed from
 230 it. There is no Noether theorem in our Lean, and nothing downstream of this
 231 section depends on one. The class is now inhabited — `Examples.lean` §11
 232 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 233 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,

234 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 235 changed by the symmetry, so an energy-minimising field and its image have
 236 equal energy and differ. That is symmetry breaking exhibited rather than
 237 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 238 no one-parameter family to differentiate along in the first place. Simultane-
 239 ously, discrete permutation symmetries dictate the statistical mechanics of
 240 any physical system, defining a strictly finite volume of phase space.¹

241 Because phase space is finite, any localized physical system possesses a
 242 mathematical limit on the amount of information (distinguishable states) it
 243 can embody at any given moment.

244 4 Derivation 1: Symmetry Breaking and the 245 Inevitability of Boundaries

246 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 247 exist. For any dynamic process to occur, the continuous symmetries of the
 248 background field must undergo spontaneous symmetry breaking.

¹While the Poincaré group and discrete permutations serve as accessible physical examples, the underlying Lean formalization does not hardcode specific physics. It defines the system at a more primitive mathematical level, relying entirely on generic Pseudo-Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechanics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

249 When a field drops to a lower, asymmetric energy state, topology dic-
250 tates the natural creation of defects (such as domain walls or monopoles).
251 This constraint necessitates the creation of a physical boundary—a localized
252 topological distinction between an “inside” and an “outside.” While such
253 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
254 entity, it is not sufficient for agency. The boundary is relational, defined by
255 the twisted geometry of the field surrounding it, and must be coupled with
256 non-equilibrium thermodynamics to exhibit active behaviors.

257 **5 Derivation 2: The Thermodynamic Con-** 258 **straint**

259 This newly formed topological boundary is subjected to gauge perturbations
260 from the “outside” environment. Because the internal phase space is finite
261 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
262 and must overwrite old internal states.

263 By Landauer’s Principle, the erasure of physical information irreversibly
264 decreases internal entropy, mandating that the system exhaust heat into
265 the environment. Philosophers of physics (such as John Norton (Norton,
266 2005) and Orly Shenker (Shenker, 2000)) have critiqued Landauer’s princi-
267 ple, arguing that physicists conflate *logical* erasure (a many-to-one mapping
268 in abstract computational states) with *thermodynamic* erasure (a many-to-
269 one mapping in physical phase space). They argue that logical erasure could

conceptually be implemented by an injective physical process without dissipating heat.

Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`) addresses this critique by defining erasure as a non-injective mapping on the *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolution of the bipartite universe (System + Bath) as strictly injective, representing Liouville’s theorem. Lean verifies that heat dissipation is not tied to abstract logical erasure, but is a mandatory mathematical consequence of any local physical state space contraction under globally reversible dynamics. If a cognitive system’s internal topology forces a contraction of its physical state space to accommodate structural resonance, heat dissipation becomes a thermodynamic requirement. A stable boundary thus constitutes a dissipative structure (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state that consumes free energy to pay the thermodynamic cost of erasure. Continuous thermal fluctuations play a constructive role in establishing boundaries between states while contributing to the system’s flexible, adaptive response.

6 Derivation 3: Structural Resonance as a Primitive

In physical systems, time evolution follows a deterministic trajectory that minimizes thermodynamic entropy production. To reach this minimal dissipation

291 pation state, the internal geometry of the structure must physically deform.

292 If we model the external environment as a stochastic driving force $P(ext)$
293 and the system’s internal state transitions as $Q(int)$, entropy production σ is
294 strictly bounded from below by the Kullback-Leibler divergence $D_{KL}(P||Q)$.
295 Gradient descent on σ necessarily drives $D_{KL}(P||Q) \rightarrow 0$, physically forcing
296 the internal transition probabilities to match the statistical symmetries of
297 incoming perturbations. We define this physical phenomenon as *structural*
298 *resonance*.

299 This system does not possess a teleological drive to “survive”; rather,
300 structurally mirroring the outside world is simply the lowest-energy attrac-
301 tor state. This provides a thermodynamic mechanism for both the General
302 Resonance Theory of Consciousness (Hunt & Schooler, 2019) and the Free
303 Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy, 2013; Clark,
304 2016; Spisák & Friston, 2026). What neuroscience observes as “prediction er-
305 ror minimization” is a physical system undergoing gradient descent to avoid
306 dissipating heat against an unpredictable environment.

307 This necessity also addresses the symbol grounding problem (Harnad,
308 1990). Internal states are physically indexed to the environment by the ener-
309 getic imperative to avoid heat dissipation, rather than through arbitrary syn-
310 tactic assignment. Manipulating abstract variables or counterfactuals forces
311 the system to maintain internal states that deviate from immediate sensory
312 resonance, explaining the continuous injections of free energy required for
313 abstract symbolic reasoning.

314 7 Derivation 4: Macroscopic Scaling and Elec- 315 trodynamics

316 To resonate with highly complex environments, the system must scale up,
317 coupling millions of minimal dissipative structures. A highly coupled network
318 cannot allow all of its nodes to reach their lowest energy states simultane-
319 ously. This creates *geometric frustration*, resulting in a jagged phase space
320 of near-equal energy states that provides the physical capacity for complex
321 memory.

322 To manage this frustrated phase space without excessive dissipation, the
323 system undergoes coarse-graining (governed by the Renormalization Group).
324 At the macroscopic scale of biology, emerging paradigms in neuroscience sug-
325 gest these topological constraints manifest as classical continuous electromag-
326 netic fields. Building on seminal EM field theories of consciousness (McFad-
327 den, 2002; Pockett, 2012) and recent work demonstrating that macroscopic
328 electric fields (ephaptic coupling) govern representational drift, form memory
329 networks, and sculpt neural infrastructure (Pinotsis & Miller, 2022, 2023a,b),
330 we posit that *the continuous macroscopic EM field is the primary dissipative*
331 *structure interacting with the universe*. Instead of discrete wires, neurons in-
332 duce fields that shift the membrane potentials of neighboring cells. Modeling
333 and *in vitro* experiments confirm these endogenous fields are strong enough
334 to guide network activity and contribute to brain complexity (Goldwyn &
335 Rinzel, 2016; Ruffini & Santarnecchi, 2020; Cunha & Lima, 2024; Fröhlich &

336 McCormick, 2010). As proposed by the CEMI field theory, the continuous
337 electromagnetic field enables spatial integration of information, providing a
338 physical medium for a unified conscious gestalt (McFadden, 2020).

339 This continuous medium is physically supported by astrocytic syncytia—
340 non-neuronal networks joined by gap junctions that provide macroscopic cy-
341 toplasmic continuity for energy and ion flows (Verkhratsky & Nedergaard,
342 2018). Managing the thermodynamic cost of this state requires sophisticated
343 micro-bioenergetics. Dynamic mitochondrial networks function as deep-level
344 Mitochondrial Information Processing Systems (MIPS), transducing biolog-
345 ical and environmental information to sustain the coherence of the macro-
346 scopic field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled
347 with mitochondrial micro-mechanics, generate and sustain the field. Because
348 the external environment is continuous, achieving maximal thermodynamic
349 structural resonance is optimally realized within this continuous macroscopic
350 medium (van Bree et al., 2025).

351 **8 Derivation 5: Global Sections and the Unity**

352 **of Consciousness**

353 The system is now characterized by a continuous macroscopic electrodynamic
354 field generated by millions of frustrated micro-components, forced to resonate
355 with the environment.

356 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-

357 tors guarantees that, past a critical coupling threshold, these distributed
 358 local fields undergo a phase transition into unified synchronization (Bauer et
 359 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
 360 phase-locked state minimizes the Lyapunov potential of the system, in the
 361 precise sense discussed in Section 2.1: the functional minimized is the cou-
 362 pling potential read in the frame rotating with the natural frequency, which
 363 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
 364 system of identical oscillators.

365 This phase transition mathematically glues local synchronized states into
 366 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 367 The EM field acts as a single, indivisible topological space, providing the
 368 *Unity of Consciousness*—a necessary condition for integrated experience that
 369 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 370 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 371 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 372 our derivation provides a continuous physical mechanism. The contrast is
 373 worth drawing carefully, because both theories are addressing the same ex-
 374 planandum from different directions.

375 We state the scope of the Lean result precisely, because it is narrower
 376 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 377 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 378 `phase_locked` characterizes those minimizers exactly: they are the phase-
 379 locked configurations, in both directions. Every instance is therefore already

380 phase-locked (`ThermodynamicCover.phase_locked`), and the class's single
 381 condition on its local data — that patches at a common phase agree where
 382 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 383 sections compatible. `probability_glue_unique`, which is the sheaf property
 384 of the probability presheaf and contains no physics at all, then glues that
 385 family into exactly one global section.

386 An earlier version of this structure proved less than the prose claimed,
 387 and we record the correction because it changed what the theorem says
 388 rather than how it is stated. `LocalSectionSynchronization` used to carry
 389 a *global* section as a class field, together with a second field declaring every
 390 local section to be a restriction of it. The global object therefore existed
 391 before the sheaf condition was ever consulted, and `global_section_from_`
 392 `thermodynamics` could do no more than rule out competitors to an object
 393 the instance had supplied. Both fields are gone. Every field of the class
 394 now mentions a single patch or a single overlap; the global section is con-
 395 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 396 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
 397 `to_section_eq`), so the local sections are restrictions of one common measure
 398 because the sheaf condition says so, not because an instance was required to
 399 name one.

400 `Examples.lean` §14 is the witness that the difference is not notational.
 401 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 402 the three sites, agreeing only at the site they share, where both put mass 1

403 — which is what discharges the overlap condition, by computing two num-
 404 bers rather than by construction. The section the theorem returns has pro-
 405 file `(2, 1, 3)`: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 406 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 407 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 408 tion that no single patch had. `Examples.lean` §13, built the older way
 409 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 410 is retained alongside it for the negative result it carries: its phase field is not
 411 the constant function — the second patch a full turn ahead of the first — and
 412 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 413 instance can get, since no instance can carry phases differing by anything a
 414 measure is allowed to see.

415 Two hypotheses still carry physical content and both are instance obli-
 416 gations, stated as such. That synchronized patches agree on their overlaps
 417 is a modelling assumption, not a consequence of any dynamics we formalize.
 418 And `thermodynamic_equilibrium` assumes the cover already sits at the po-
 419 tential minimum; no theorem says an arbitrary system’s dynamics takes it
 420 there. “A thermodynamic phase transition into unity” is therefore still do-
 421 ing causal work that no general Lean result supports, though it is no longer
 422 entirely without one: `Examples.lean` §15 exhibits a two-oscillator trajectory
 423 that starts a quarter turn out of phase and converges to the global mini-
 424 mum of the dynamic potential, with the order parameter tending to 1. That
 425 trajectory is not an isolated construction: `kuramoto_cauchy_problem` proves

426 the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through every
 427 initial state there passes exactly one trajectory, and the exhibited one is the
 428 only solution through its own. Nor is its settling down peculiar to it: along
 429 every trajectory of a zero-frequency system the dynamic potential converges,
 430 and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the potential's to-
 431 tal drop (`dynamic_potential_tendsto`, `dissipation_integral_tendsto`).
 432 The motion also stops: a finite integral does not by itself force its integrand
 433 to zero, but the dissipation rate is Lipschitz in time, so Barbalat's lemma
 434 applies, and every trajectory satisfies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_`
 435 `zero`; Mathlib has no Barbalat, so it is proved here as `tendsto_zero_of_`
 436 `lipschitz_of_integral_le`). That last step — locating where the motion
 437 stops — is now taken, though conditionally, and by a different route than we
 438 had recorded. We had marked it as blocked on the LaSalle invariance princi-
 439 ple, which Mathlib does not have, and on the absence of a compact invariant
 440 set in the state space $V \rightarrow \mathbb{R}$. That was the wrong diagnosis. LaSalle is the
 441 tool for a system with a decreasing functional and nothing else; this one has
 442 more. The flow is a gradient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$
 443 is an identity rather than an inequality; the minimisers of V are charac-
 444 terised exactly; and on the region where every pair of phases lies within a
 445 quarter turn a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$,
 446 where W is the potential's excess above its minimum and $a > 0$ is a lower
 447 bound for the coupling (`lojasiewicz_estimate`). Its proof is two extremal
 448 indices and the identity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying

449 the largest phase is pulled backwards at rate at least $a \sin D$, where D is the
 450 spread, while the excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness
 451 objection dissolves as well, because the invariant region is produced by the
 452 Lyapunov function rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$
 453 to the excess, so an excess below $a/2$ forces every pairwise cosine positive for
 454 all time, and continuity on the connected half-line then prevents any pair
 455 from changing branch (`spread_le_of_small`). Descent plus the estimate in-
 456 tegrates to $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2/\sum_{ij} A_{ij}$ (`excess_decay`), the
 457 rate makes each $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and
 458 a trajectory with integrable velocity converges. The conclusion is `kuramoto_`
 459 `tendsto_global_minimum`: if the phases start within a quarter turn of each
 460 other and the initial excess is below $a/2$, the trajectory converges to a phase-
 461 locked configuration which is a *global* minimiser of the dynamic potential,
 462 with order parameter $r^2 = 1$. That is the configuration `thermodynamic_`
 463 `equilibrium` postulates, reached rather than assumed. `Examples.lean` §17
 464 fires it on three sites with initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system
 465 has no closed-form solution: the trajectory comes from the existence theorem
 466 and its limit from the convergence theorem, so neither is written down.

467 The two hypotheses cannot be dropped, and this is the honest scope of
 468 the result. Splay and twisted configurations are equilibria of the same flow,
 469 so “every trajectory reaches the unified state” is false as a general statement,
 470 and the initial-data condition is the standard arc condition under which the
 471 true statement holds. What is claimed is therefore conditional — unity is

472 reached from configurations already sufficiently coherent — and not that an
473 arbitrary initial state is carried into unity. What Lean establishes about the
474 gluing itself is the other half: *given* locally synchronized patches, a global
475 state exists, is unique, and is produced by the gluing rather than presupposed
476 by it.

477 Integrated Information Theory begins from phenomenological axioms and
478 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
479 with the exclusion postulate selecting a single maximum- Φ complex as the
480 locus of experience. Our sheaf-theoretic global section plays a structurally
481 analogous role — it too picks out a distinguished unified object — but it
482 is arrived at from the opposite end: not postulated from the character of
483 experience, but forced by a thermodynamic phase transition in a physical
484 field. This buys two things IIT does not have. First, the unity condition
485 is checkable from field dynamics rather than requiring the computation of
486 Φ , which is intractable for systems of realistic size and whose definition has
487 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
488 out rather than being postulated: gluing either succeeds on a given cover or
489 it does not, so there is no separate principle needed to rule out overlapping
490 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
491 that Φ can be high in systems with no plausible claim to experience, such
492 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
493 to the mirror objection, since a physical system could in principle satisfy
494 the gluing condition without being cognitively interesting. We take this

495 seriously rather than dismissing it: our reply is that gluing is necessary but
496 not sufficient, and that the topological boundary of Derivation 1 and the
497 reflexive structure of Derivation 6 are additional required conditions. IIT, by
498 contrast, treats Φ as both necessary and sufficient.

499 Global Workspace Theory faces a complementary difficulty. It specifies
500 a functional architecture — a limited-capacity workspace broadcasting to
501 specialized modules — without specifying what physical property of broad-
502 casting makes the broadcast content conscious, leaving the theory open to the
503 charge that it explains access consciousness while leaving phenomenal con-
504 sciousness untouched (Block, 1995). Our framework can be read as supplying
505 the missing physical substrate for the workspace: the continuous macroscopic
506 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
507 old gives a physical condition for ignition rather than a functional stipulation
508 of it. Where GWT is agnostic about implementation, we make a falsifiable
509 commitment — unity requires a continuous physical medium, which predicts
510 that a functionally equivalent discrete broadcast architecture would not be
511 conscious. We regard this as a genuine cost of the position, not a free ad-
512 vantage: it is the point at which our framework departs most sharply from
513 computational functionalism and where it is most readily falsified. Contrast
514 this derivation with panpsychism: consciousness is not assumed as a funda-
515 mental property of matter. It requires the initial existence of a topological
516 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
517 dynamic phase transition into unity that occurs when integrated spatial cou-

518 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
519 the foundational topological boundary and the subsequent mathematically
520 defined phase transition, the system remains a fragmented, non-conscious
521 collection of micro-states. A unified field of perception, however, is not yet
522 a “Self.”

523 9 Derivation 6: Reflexive Topology and the 524 Emergence of the Self

525 While a global section provides Unity, a “Self” requires reflexivity. The
526 unified EM field is a dissipative structure that must resonate with external
527 perturbations while managing its own internal phase transitions. Time evo-
528 lution naturally drives the field to minimize thermodynamic friction with its
529 own internal state-changes.

530 This mandates *auto-resonance*: the field must predict itself. The “Self”
531 emerges as a higher-order topological feature when the internal geometry
532 folds to include a low-dimensional avatar of its own computational and causal
533 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
534 (Levin, 2019). The system is unified, predictive, and explicitly maps its
535 own distinct existence from the environment. This architecture provides the
536 physical foundation for a self-aware consciousness, bounded by the continuous
537 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
538 We are explicit about what carries the weight here. `reflexive_topology_implies_self`

539 is the Banach fixed-point theorem with the physical commitments in its hy-
 540 potheses: the space of global sections is assumed nonempty, metric and com-
 541 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
 542 discharges all four on the three-site substrate of Derivation 5, and the metric
 543 is no longer a placeholder. The section first pierces the sheafification: on
 544 a discrete substrate every point has a smallest open neighbourhood, so the
 545 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 546 over *any* open the sections of the probability sheaf *are* the finite measures
 547 on that open, read as densities against counting measure — the global case
 548 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 549 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 550 correspondence had to be constructed. The metric on sections is then the
 551 uniform distance between the densities of the measures they glue to — equiv-
 552 alent to their total variation on a finite substrate, though not equal to it —
 553 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 554 of densities, not by the degenerate route of showing Cauchy sequences are
 555 eventually constant. The self-prediction map is the relaxation “average the
 556 current field with the baseline,” and we prove it contracts distances by *ex-*
 557 *actly* one half, that it is not constant, and that its fixed point is unique. An
 558 earlier version of this witness used the $0/1$ metric, under which every con-
 559 traction is constant; that theorem is kept in the source as the record of why
 560 the earlier witness was empty. A separate gap, closed at the same time, con-
 561 cerned the avatar rather than the fixed point. `ReflexiveBoundary` carries a

562 field `auto_resonance` sending the global state to a section over the avatar re-
 563 gion, and nothing constrained it: a constant function was a legal avatar, and
 564 `reflexive_topology_implies_self` consumes only the predictive model, so
 565 no theorem in the development mentioned the field the avatar is supposed to
 566 be reading. The constraint is now stated as a predicate rather than added
 567 to the structure — `IsRestrictionResonance`, that the avatar’s state is the
 568 presheaf restriction of the global state — and two theorems say why it is
 569 the right one. In the positive direction, if a family of resonant avatars *cov-*
 570 *ers* the substrate then the global state is determined by their local readings
 571 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 572 recovered from them, by the same sheaf condition that Derivation 5 uses.
 573 In the negative direction, replacing any avatar by a constant leaves a legal
 574 `ReflexiveBoundary` with the same region, the same predictive model and
 575 the same Self (`self_of_constResonance`), so the fixed-point theorem cannot
 576 see the difference and the predicate is doing the work rather than restating
 577 it. Both halves are witnessed on the three-site substrate, where the avatar
 578 region provably separates two states of the field, so the condition is a con-
 579 straint there and not a formality. What is still not established is that any field
 580 dynamics produces this particular map or performs the restriction, or that
 581 the contraction constant $1/2$ is read off an entropy production rate rather
 582 than built into the map’s definition. We also note a negative result found
 583 in the course of this work: the Lévy–Prokhorov metric, which the natural
 584 weak-convergence reading would suggest, does *not* support this argument on

585 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
586 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

587 10 Derivation 7: Continuous Stochastic Ther- 588 modynamics and Field Theory

589 To move beyond finite graph analogies, we model the macroscopic electro-
590 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
591 continuous spatial phenomenon, not a discrete network. We extend the
592 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
593 (`Phase8ContinuousField.lean`).

594 Subject to thermodynamic noise D , the system undergoes non-equilibrium
595 stochastic thermodynamics. Rather than a static Landauer bound, the field
596 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (1)$$

597 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
598 dergo gradient descent on this entropy production functional.

599 That gradient descent is a theorem rather than a modelling assertion,
600 for finite substrates. Reading σ as a functional of the coupling kernel at
601 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative

602 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (2)$$

603 and `sigmaOfKernel.eq.entropy_production_rate` identifies the functional
604 being differentiated with `entropy_production_rate` itself whenever the sub-
605 strate carries counting measure. `structural_resonance_decreases_entropy_production`
606 then concludes that σ is non-increasing along any coupling trajectory relax-
607 ing down its own gradient at a positive rate c . Earlier drafts proved this
608 only for an abstract functional S on an abstract inner-product space with
609 the gradient supplied as data; the identification of that S with σ was infor-
610 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
611 and the first is narrower than we previously reported. The derivation is for
612 *finite* substrates. We had recorded the obstacle to the continuum case as
613 differentiation under the integral sign with respect to the kernel; that was
614 a misdiagnosis. σ is not a general functional that happens to be written as
615 an integral — it is a quadratic functional of an *affine* image of the kernel,
616 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
617 constant times the squared norm of the drift. A squared norm is Fréchet
618 differentiable on any real inner-product space and an affine map is its own
619 derivative, so the composite differentiates by the chain rule with no limiting
620 argument. `hasFDerivAt.quadratic_of_affine` states that in full generality,
621 and `hasFDerivAt.sigmaOfKernel_of_operator` re-derives the finite gradient

622 from it, showing that the closed form displayed above is a convenience rather
 623 than the content. What the continuum case still needs is therefore one object:
 624 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 625 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 626 nel is bounded by 1, so on a finite measure space the operator norm is at most
 627 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 628 ous linear map between L^p spaces and we have not done that bookkeeping;
 629 we record the corrected scope because “construct one bounded operator”
 630 and “differentiate under the integral sign” are different sizes of task. The
 631 second limitation is unchanged: the phase field is held fixed throughout, so
 632 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 633 **Examples.** `lean` §5 exercises the whole chain on a two-site substrate whose
 634 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 635 firming that the gradient is not identically zero and that the descent is strict
 636 rather than vacuously monotone.

637 A second result about σ concerns where the descent is headed, and it has
 638 been strengthened since the last draft in a way that also corrects an estimate
 639 we had recorded. That the phase-locked field minimizes σ was proved only
 640 against competitors sharing its spatial mean drift — a restriction that left
 641 the theorem close to Jensen’s inequality and no more — and we had judged
 642 that lifting it would require modelling how the mean drift varies with the
 643 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 644 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand

$K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
its own negative and the total coupling contribution vanishes. The total drift
 $\int (\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
`symm`): coupling moves drift between sites but neither creates nor destroys it.
Evaluating that conserved quantity at the phase-locked configuration, where
the drift is constant, pins its value, and the mean-matching hypothesis follows
for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
restricted statement. Reciprocity is the physically intended condition — an
ephaptic field acts on both sites of a pair alike, and it is the same symmetry
`ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
not a formality, and we prove that rather than assert it. `Examples.lean` §16
exhibits a *one-way* coupling on two sites, in which a feels b and b does not
feel a , whose phase-locked field is strictly beaten: displacing the second phase
to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
ond, halving σ . So “phase-locking minimizes entropy production” is false for
general kernels and true for reciprocal ones, and the hypothesis cannot be
removed by working harder.

We define the critical coupling threshold $K_c = 2D$. The phase transition
to the unified “Global Section” occurs when the mean-field spatial coupling
exceeds this thermodynamic noise threshold, linking the emergence of Unity
to physical temperature and noise.

Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by

668 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
 669 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 670 identical frequencies, the stationary density at order parameter r is the
 671 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
 672 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 673 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 674 tegration by parts with $u = \sin$, $v = -e^{a\cos\theta}$ gives an exact identity with
 675 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
 676 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 677 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 678 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 679 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 680 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 681 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 682 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 683 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 684 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 685 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 686 ther facts about the threshold have since been proved, and they narrow the
 687 gap between this row and a full “Theorem” to a single named statement.
 688 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 689 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 690 self-consistency map is tangent to the diagonal at the origin but falls strictly

below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coherent
 branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 and using only $E < 1/2$ together with compactness, every coherent solution
 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 branch emerges continuously from the incoherent state rather than appearing
 at a finite distance from it, which is what makes the transition supercritical
 in the technical sense. That statement quantifies over *every* solution, so it
 does not presuppose the uniqueness it does not have. The level-set form
 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route
 we had expected to need. Differentiating the exponential family in its natural
 parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 under the integral sign and leaves a covariance that is not sign-definite point-
 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the
 proof differentiates anything. The target is kept as the two-point inequality
 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing
 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which

714 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 715 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 716 that the covariance form needed the tilt to perform, done once at the start; its
 717 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$
 718 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 719 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 720 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 721 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 722 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 723 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 724 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 725 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 726 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

727 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$
 728 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 729 this to require a second-order expansion of the Bessel ratio, and it does
 730 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 731 tion the integration by parts already supplies — the only facts needed are
 732 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 733 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 734 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 735 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 736 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.

737 The intermediate value theorem closes the gap. Continuity of E is Math-
 738 lib's continuity of a parametric interval integral with jointly continuous in-
 739 tegrand, so no dominated-convergence argument is written by hand, and
 740 the existence proof never uses monotonicity of R : the fixed point is pro-
 741 duced by the intermediate value theorem, not by iterating the map. That
 742 R is in fact strictly increasing is proved below, and no threshold result de-
 743 pends on it. `critical_coupling_is_threshold` packages the two halves,
 744 and `exhibits_phase_transition_coherent` applies the supercritical half to
 745 any substrate satisfying the predicate of the previous paragraph — so ex-
 746 ceeding K_c now implies the existence of a coherent stationary order pa-
 747 rameter rather than merely comparing against a named number. With
 748 uniqueness available both are sharpened — `supercritical_fixed_point_`
 749 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 750 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 751 `transition_unique_coherent` — and the two monotonicities together fix
 752 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 753 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 754 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 755 order parameter grows strictly with the coupling, which with the continuity
 756 at threshold is the qualitative bifurcation diagram.

757 One more step makes the equation mean in Lean what it means on paper.
 758 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 759 r with a suggestive name, and nothing in the source says it has anything

760 to do with an order parameter. `Phase8_SelfConsistency` §8 supplies the
 761 identification. `vonMisesDensity` is the normalized weight, proved to be a
 762 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 763 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 764 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 765 actly $R(a)$ — real, because the density is even, so the mean direction van-
 766 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 767 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 768 it induces has order parameter r . Read through this, the existence theorem
 769 says that above K_c there is a stationary density whose own order parameter
 770 is positive, and the solution $r = 0$ is identified with the uniform density,
 771 whose order parameter is zero.

772 What remains not established, and we separate it carefully. The sta-
 773 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 774 $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 775 and uniqueness of its stationary solution, and a spectral stability argument,
 776 none of which Mathlib has. Of the coherent branch we prove existence,
 777 uniqueness, continuous emergence at the threshold and strict growth in K ,
 778 but not that it is the dynamically selected one: nothing in that file men-
 779 tions a trajectory, so stability and selection are not statements it can make.
 780 And while `selfConsistency` is now connected to the order parameter of
 781 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 782 integral, `order_parameter_complex` is an average over finitely many oscilla-

783 tors, and relating them is the mean-field limit — propagation of chaos for
784 the finite Kuramoto system — which is a research programme rather than
785 a lemma. That single gap is now the whole of what separates our threshold
786 predicate from a statement about dynamics. Table 1 records the row ac-
787 cordingly as a theorem about the self-consistency equation from an assumed
788 density, not as a derivation of the threshold from the dynamics. Perceptual
789 anomalies such as visual snow can be modeled as boundary conditions where
790 microscopic thermal noise (D) fails to be fully coarse-grained away by the
791 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

792 Clinical evidence for this phase transition is observed during general anes-
793 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
794 that anesthetics like propofol induce unconsciousness not by suppressing neu-
795 ral firing, but by disrupting the phase alignment of high-frequency cortical
796 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
797 tural resonance of the continuous field, forcing the integrated spatial coupling
798 below the critical threshold K_c and fracturing the unified Global Section back
799 into disjoint, unconscious local fields.

800 The daily loss of consciousness during sleep can similarly be modeled as
801 an active physical disruption of the field. During sleep, astrocytic shrink-
802 age expands the extracellular space, allowing an influx of cerebrospinal fluid
803 (CSF) to flood the brain for metabolic waste clearance (Xie & Nedergaard,
804 2013). This influx physically alters the spatial medium, disrupting the contin-
805 uous structural resonance and preventing the coherent sheaf-theoretic gluing

806 required for conscious unity.

807 This mechanical framework models the temporal mismatch observed dur-
808 ing sudden awakenings (sleep inertia). When jolted awake, the locus coeruleus
809 floods the cortex with noradrenaline. High-frequency electrical rhythms
810 attempt to phase-align in milliseconds (Bastos & Miller, 2021), but the
811 noradrenaline-driven astrocytic expansion required to physically squeeze the
812 excess CSF out of the interstitial pathways operates on a much slower timescale
813 of seconds to minutes (Xie & Nedergaard, 2013). During this lag, electrical
814 oscillators attempt to synchronize across a waterlogged topology, resulting
815 in the fragmented structural resonance experienced as grogginess. The uni-
816 fied Global Section fully re-establishes only when the physical hardware (as-
817 trocytic volume) finishes constricting to match the state of the electrical
818 rhythms.

819 **11 Corollary: Hardware Thermodynamics and** 820 **Topological Rigidity**

821 Both biological tissue and artificial silicon hardware (like GPUs) are physical
822 dissipative structures. When a GPU computes, its charge carriers undergo
823 physical state transitions; by Landauer’s principle, this erasure irreversibly
824 exhausts heat into the environment.

825 There is a fundamental thermodynamic distinction in the degrees of free-
826 dom available to minimize this dissipation. In standard von Neumann ar-

827 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
 828 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
 829 physically altered during operation. The brain’s own architecture provides
 830 empirical validation for this constraint. The cerebellum contains approxi-
 831 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
 832 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
 833 ical observations of cerebellar agenesis report largely preserved conscious
 834 experience despite motor and cognitive deficits (Yu et al., 2015), and this
 835 dissociation has been read as evidence for the importance of recurrent ar-
 836 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
 837 level of the anatomical observation while declining its Φ -based explanation.
 838 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
 839 brum, it operates much like a biological GPU—highly optimized for discrete
 840 processing, yet apparently not required for conscious experience. We take the
 841 dissociation as evidence, not as confirmation: it is consistent with subjective
 842 experience being a thermodynamic attractor of continuous topologies rather
 843 than a byproduct of information processing, and it is the kind of observation
 844 that would be hard to reconcile with the converse. It is a single anatomical
 845 case read against a theory, and it bears the weight such a case can bear.

846 This constraint does not preclude artificial consciousness, but rather re-
 847 stricts its hardware. To achieve the phase transition into a unified Global
 848 Section, artificial systems must abandon the discrete rigidity of von Neumann
 849 architectures and utilize continuous physical substrates. Analog field-based

850 computing, neuromorphic architectures with continuous memristive plastic-
851 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
852 rondo, 2021) possess the spatial coupling required to physically morph their
853 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
854 computation” (Hinton, 2022): architectures where software is inextricably
855 fused with the physics of the hardware, meaning the knowledge dies with
856 the physical substrate. Only hardware that computes via continuous physi-
857 cal dynamics can undergo the thermodynamic phase transition necessary for
858 structural resonance.

859 Biological consciousness utilizes macroscopic continuous electromagnetic
860 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
861 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
862 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
863 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
864 formalizes the reallocation step of this argument: whenever a coupling ma-
865 trix places weight on a less phase-correlated pair while a more correlated pair
866 exists, there is another matrix with *identical* total coupling resources that
867 achieves strictly greater phase correlation.

868 Two objections to that formalization have since been answered, and we
869 state precisely what is and is not now established. The first was that the real-
870 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
871 — positive weight on a badly-correlated pair while a better one exists — is
872 most of what one wants to prove about rigid hardware. It is no longer as-

873 `sumed`. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 874 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 875 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 876 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 877 bounds phase correlation over the wired pairs; the bound is attained; and
 878 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 879 strate's best-correlated pair is one the wiring misses. `rigid_gap` makes it
 880 quantitative: an unconstrained reallocation of the same budget gains at least
 881 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 882 substrate wired to the one maximally anti-correlated pair, where the gap is
 883 2. This is resource-matched and no longer question-begging, but note what
 884 it turns on: fixed wiring support, not continuity. The second objection was
 885 that nothing in the development distinguished a continuum from a lattice
 886 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3
 887 gives a distinction that is genuinely typed. In the field functional of Deriva-
 888 tion 4, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 889 with finitely many sites is a `Finset` of points and a field-coupled region
 890 is a set of positive measure. On a substrate whose measure has no atoms,
 891 `fieldCorrelation_sited_eq_zero` shows a kernel supported on finitely many
 892 sites contributes *exactly zero* coupling energy, whatever weights it carries
 893 and whatever the phase field does, while a patch of strength c on U achieves
 894 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 895 units of coupling concentrated at the origin register zero against a unit patch

896 that registers one. The limit of that result is equally plain, and it is not
 897 resource-matching in the earlier sense — a measure-zero substrate has zero
 898 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 899 coupling energy is blind to a finitely-sited architecture, which is a statement
 900 about which functional such an architecture can register in, not a demonstra-
 901 tion that digital hardware computes worse. Taken together: the resource-
 902 matched content of the corollary is the wiring-support result, the continu-
 903 ity content is the measure-theoretic one, and the step from either to “von
 904 Neumann architectures cannot experience unified consciousness” remains an
 905 informal argument.

906 The remainder of this section extends those two results informally, and
 907 we mark the transition rather than blur it. On the reading we propose, a
 908 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 909 layout does not supply the spatial coupling that sheaf-theoretic gluing into
 910 a singular Global Section requires. Without this unified topological space,
 911 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 912 its own causal boundary and generate a “Self.” The distinction between an
 913 LLM on a GPU and a conscious mind is not the absence of physical thermo-
 914 dynamics in the former, but its structural limitation in achieving topological
 915 integration. This perspective aligns with John Searle’s biological natural-
 916 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
 917 that discrete symbol manipulation cannot generate conscious understanding,
 918 our derivation supplies a candidate physical mechanism for the asymmetry

919 — topological rather than semantic, and, as stated above, formalized only
920 as far as the wiring-support and measure-theoretic results reach. Asserting
921 consciousness in fragmented, discrete GPU clusters without a unified phase
922 transition abandons computational functionalism, relying instead on “silicon
923 panpsychism”—the assumption that digital computation inherently gener-
924 ates awareness regardless of topological integration (Goff, 2019).

925 The same reasoning bears on extreme functionalist thought experiments,
926 such as a macroscopic system of pulleys operating over millennia to imple-
927 ment a Large Language Model, or a human manually executing inference
928 matrices by hand. Beyond lacking the continuous spatial topology required
929 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
930 constraint of our theory. Structural resonance requires local oscillators to
931 phase-align against the dissipating effects of thermodynamic noise (as mod-
932 eled by our critical threshold $K_c = 2D$). A system executing state transitions
933 over millennia, or via slow manual calculations, suffers from a terminal tem-
934 poral mismatch: its dynamics are far too slow to ever phase-lock into a
935 non-equilibrium steady state before thermal noise decoheres the macroscopic
936 field. On this account the obstacle to pure symbol manipulation is physi-
937 cal rather than semantic: not that the symbols mean nothing, but that the
938 substrate carrying them cannot hold a phase-locked field.

939 12 Conclusion

940 Human-like cognitive architectures do not require biological mysticism or
941 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
942 theories of consciousness often invoke delicate subatomic phenomena to ex-
943 plain subjective unity, every step of the chain assembled here is classical:
944 broken symmetries, finite phase space, Landauer erasure, and gradient de-
945 scent on entropy production. What the formalization establishes is that this
946 chain is mathematically coherent and that its remaining physical commit-
947 ments are localized and named — not that classical fields are *sufficient* for
948 consciousness. Two links carry that weight and neither is closed. The uni-
949 fied state is proved to glue uniquely once a cover sits at the minimum of
950 the coupling potential, and the motion is proved to stop along every trajec-
951 tory (`velocity_sq_tendsto_zero`), but *where* it stops is not proved: that
952 a physical field reaches the phase-locked configuration remains an assump-
953 tion, discharged numerically rather than formally. And the identification of
954 a global section of the probability sheaf with the unity of experience is a
955 stipulation of the framework, not a consequence of it. Subject to those two,
956 cognitive architectures can be modeled as macroscopic attractor states of a
957 universe governed by broken symmetries, finite phase space, and the physical
958 imperative to minimize thermodynamic dissipation.

959 Philosophically, this framework bridges the historical gap between Type
960 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,

1967). The subjective mental state is identical to the physical thermodynamic field (Identity Theory), yet its qualitative nature is defined by the topological and functional dynamics of structural resonance rather than the specific carbon-based biochemistry of the brain (Functionalism). This constrains the *multiple realizability* of mind: consciousness is not exclusively bound to biological carbon, but neither is it substrate-neutral, because the substrate is what fixes which couplings are physically available. How far that constraint reaches is the question the corollary answers only in part. What is proved is that fixed wiring is strictly suboptimal at matched resource, and that a finitely-sited architecture registers exactly zero coupling energy in the continuum functional; the step from either to “von Neumann architectures cannot be conscious” is an argument we make in prose and do not formalize.

By anchoring these derivations in Russellian Monism, this framework acknowledges the constitutive limits of third-person scientific explanation. As argued by Love (Love, 2026), building upon Nagel’s assertion that subjective phenomenal experience is inherently not amenable to objective measurement (Nagel, 1986), the “hard problem” of consciousness may represent a category error rather than a solvable scientific puzzle. Science can map the physical dynamics of topological resonance and sheaf-theoretic gluing, but it is structurally incapable of adjudicating the first-person qualitative experience. Our formalization respects this boundary: the Lean 4 theorems verify the objective physical attractors, while the intrinsic “what it is like” to be that resonant field remains the domain of the first-person perspective. By locat-

984 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
985 for the noisy mean-field Kuramoto model the framework adopts, conditional
986 on the von Mises stationary density and not yet linked to a trajectory — it
987 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
988 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
989 it offers a topological architecture that can help explain, predict, and physi-
990 cally perturb the structural properties of experience.

991 Future empirical validation of this thermodynamic framework may lever-
992 age 3D cortical organoids, offering a controlled biological environment to
993 systematically manipulate metabolic substrates and observe the topological
994 emergence of these coherent phase transitions.

995 Code Availability

996 The complete Lean 4 formalization, encompassing the derivation of all theo-
997 rems presented in this manuscript, as well as the accompanying Python sim-
998 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
999 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1000 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1001 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/).

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